

Project Summary

Date	13 July 2026
Project Name	Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

Project Summary

The **Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau** project was developed to analyze cosmetics sales trends, customer preferences, and product performance using modern data analytics techniques. The project transforms raw cosmetics sales data into meaningful business insights through structured database management, interactive Tableau dashboards, and a web-based visualization platform.

The dataset was collected in CSV format, cleaned, and processed using Microsoft SQL Server. SQL queries were used to perform data extraction, filtering, aggregation, and trend analysis. The processed data was connected to Tableau Public to create interactive dashboards and stories that present product category analysis, brand performance, regional sales trends, customer purchasing behavior, and revenue insights. A Flask web application was developed to integrate these dashboards into a responsive interface built with HTML, CSS, and Bootstrap.

Objectives Achieved

- Collected and prepared the cosmetics sales dataset.
- Designed and implemented a Microsoft SQL Server database.
- Performed SQL-based data analysis.
- Developed interactive Tableau dashboards and stories.
- Built a Flask web application for visualization.
- Published dashboards using Tableau Public.
- Maintained project documentation and GitHub repository.

Technologies Used

- Python
- Flask
- Microsoft SQL Server
- Tableau Public
- HTML
- CSS
- Bootstrap
- Git & GitHub

Key Outcomes

- Interactive visualization of cosmetics sales trends.
- Brand-wise and product category analysis.
- Regional sales and customer purchasing insights.
- Revenue and profit performance analysis.
- Responsive web interface for accessing dashboards and reports.

Future Enhancements

- Integrate real-time cosmetics sales data.
- Add AI-based product recommendation features.
- Implement customer segmentation and predictive analytics.
- Optimize database performance for large-scale retail datasets.
- Deploy the application on a cloud platform for public access.

Conclusion

The project successfully demonstrates how data analytics can be used to understand cosmetics market trends and consumer behavior. By combining Microsoft SQL Server, Tableau Public, Flask, and modern web technologies, the solution provides an efficient, interactive, and scalable platform for analyzing cosmetics sales data. The project supports data-driven business decisions and can be further enhanced with predictive analytics, customer personalization, and real-time reporting.